

Discovering Phase Transitions in the 2D Ising Model with Unsupervised Learning

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I. INTRODUCTION

Phase transitions are one of the most fundamental phenomena in condensed matter and statistical physics. At a phase transition, a system goes through a qualitative change in its macroscopic behavior, for example changing from ferromagnetism to paramagnetism in a magnetic material. There is a quantity that is nonzero in the ordered phase and vanishes in the disordered phase called the order parameter, however, both Wang and Wetzel [1, 2] note that identifying the correct order parameter becomes significantly harder in more complex systems, for example, topological insulators, and quantum spin Hall states, in which the order parameter may only be defined in a complicated nonlocal way, or the Hamiltonian itself is not fully known. This motivates the development of data driven methods capable of identifying phase transitions and their associated order parameters directly from raw configurations, without requiring prior physical knowledge.

In this paper, we use the two dimensional ferromagnetic Ising model, which is an ideal test case for these methods: it has an exactly known critical temperature due to Onsager [4] and well-understood finite-size behavior, providing a precise ground truth against which machine learning estimates can be validated. The model is defined on a square lattice by the Hamiltonian

$$H = -J \sum_{\langle i,j \rangle} S_i S_j \quad (S_{i,j} \in \{+1, -1\})$$

where $J > 0$ is the ferromagnetic coupling constant and S defines whether it is spin up (+1) or down (-1). The spin configurations used were generated via Monte Carlo simulation, producing 10,000 independent configurations at each of 16 temperatures spanning the ordered and disordered phases. We apply both supervised and unsupervised machine learning methods to Monte Carlo configurations of the 2D ferromagnetic Ising model. The supervised classifier consists of a convolutional neural network (CNN). These will be used to establish a performance baseline and verify that the critical temperature can be accurately located from labeled data.

The critical temperature was solved exactly by Onsager [4] as

$$T_c = \frac{2J}{k_B \ln(1 + \sqrt{2})} \approx 2.2698K/k_B$$

This provides a precise ground truth against which we can compare our machine learning estimate.

However, the primary focus is on unsupervised methods. We apply principal component analysis (PCA) follow-

ing Wang [2] to demonstrate that the magnetization order parameter can be recovered without labels, and an autoencoder following Wetzel [1] to show that the reconstruction error independently identifies T_c . We additionally apply t-distributed stochastic neighbor embedding (t-SNE) as a nonlinear visualization tool following Wetzel [1], though the primary unsupervised results are drawn from PCA and the autoencoder methods. Since the square lattice Ising model has a linear order parameter, PCA is expected to perform well here. The supervised and unsupervised results together serve as a controlled verification of these methods before they would be applied to more complex systems where no exact solution exists.

II. METHODS

The methods used in this paper consist of both supervised and unsupervised learning, with the primary focus on the unsupervised side. Before getting into the models themselves, it is worth describing the data and its structure.

The dataset consists of spin configurations of the 2D ferromagnetic Ising model on a square lattice with $L = 40$, giving $N = 1600$ spins per configuration. Temperatures range from $T = 0.25$ to $T = 4.00$ in steps of 0.25, covering both the ordered and disordered phases around T_c . Each temperature has 10,000 configurations, giving 160,000 total samples. Configurations are stored as packed bits, which were unpacked and remapped from $\{0, 1\}$ to $\{-1, +1\}$ so that they correspond to physical spin values.

Following Wang [2], Principal Component Analysis (PCA) is applied directly to raw spin configurations without any labels. PCA is a feature extraction technique that finds a linear transformation $Y = XP$ where the columns of P are orthogonal weight vectors called principal components [5], determined by the eigenproblem

$$X^T X p_l = \lambda_l p_l$$

[5] The eigenvalues λ_l , when normalized by their sum, give the explained variance ratio. The first principal component captures the direction of greatest variance in the data. PCA was fit on a random subsample of 25,000 configurations and then applied to the full dataset. We retain 8 principal components, though primarily the first two are used. The critical temperature is estimated from $\langle |p_1| \rangle$ by finding the temperature of the steepest descent.

k-means clustering with $k = 3$ is applied to the two-dimensional space spanned by PC1 and PC2 on a subsample of 5,000 points, following Wang [2]. The choice $k = 3$ is

physically motivated by the expected phase structure: two ferromagnetic ground states corresponding to positive and negative magnetization, and one paramagnetic state centered near zero magnetization. This is used to visualize how the ferromagnetic and paramagnetic states separate in the reduced feature space.

Following Wetzel [1], two-component t-distributed stochastic neighbor embedding (t-SNE) is applied as a nonlinear visualization method. To keep computation manageable, PCA is first used to project configurations to 50 dimensions, and t-SNE is then run on a subsample of 5,000 configurations.

An Autoencoder (AE) is implemented in PyTorch. Unlike PCA, it is no longer restricted to linear transformation, the encoder compresses the input through a bottleneck and the decoder attempts to reconstruct it, forcing the network to learn a compact representation [1, 3]. The encoder maps from 1600 to 2 with ReLU activations, producing a 2D latent vector Z . The decoder mirrors this in reverse and uses tanh on the output layer to match the ± 1 spin domain. The network is trained on 25,000 configurations for 50 epochs using the Adam optimizer with a step learning-rate scheduler, minimizing mean-squared-error (MSE) loss. T_c is estimated from the latent coordinate Z_1 and from the per-temperature reconstruction error, both via their steepest-gradient temperature.

A Variational Autoencoder (VAE) with a 1D latent space is also implemented, using the same encoder/decoder architecture as the AE. As described by Wetzel [1], the key difference is that the encoder outputs a mean μ and log-variance, from which the latent variable is sampled via the reparameterization trick. The total training loss is the sum of the MSE reconstruction term and a Kullback-Leibler divergence regularizer. Like the AE, the VAE is trained on 25,000 configurations for 40 epochs with Adam. T_c is estimated from both the reconstruction error curve and μ .

Lastly, for the supervised baseline, a Convolutional Neural Network (CNN) is trained to classify each spin configuration as ferromagnetic or paramagnetic, labeled 0 and 1 respectively. Configurations are reshaped to 40×40 images. The architecture consists of three convolutional blocks followed by fully connected layers, with dropout to reduce overfitting. T_c is estimated as the temperature where the mean predicted ferromagnetic probability crosses 0.5.

III. RESULTS

A. Observables

From the data, we first confirm that the Monte Carlo simulations reproduce the known physics of the 2D Ising model. Figure 1 shows the spin configurations at each temperature. The ferromagnetic (FM) phase is represented by the solid yellow colour, and it is clear that from $T = 0.25$ to $T = 1.50$ the lattice is nearly fully aligned. Near the critical temperature at $T = 2.25$, large fluctuating domains appear and the lattice

begins to disorder. Above T_c , the spins are fully disordered, characteristic of the paramagnetic (PM) phase.

Figure 2 shows the magnetization distributions at three representative temperatures. At $T = 0.50$, the distribution is sharply peaked at $m \approx \pm 1$, confirming strong ferromagnetic order. Near T_c at $T = 2.25$, the distribution broadens and develops a bimodal structure, reflecting the large fluctuations at the critical point. At $T = 4.00$, the distribution collapses to a single peak at $m \approx 0$, confirming the disordered paramagnetic phase.

Figure 3 shows the order parameter $\langle |m| \rangle$ and the magnetic susceptibility χ as functions of temperature. The order parameter drops sharply near the exact $T_c = 2.2692$, while the susceptibility peaks at $T = 2.25$, within one temperature step of the exact value. These results are consistent with known finite-size behavior at $L = 40$ and serve as a reliable ground truth against which the machine learning estimates will be compared.

B. Principal Component Analysis (PCA)

PCA is applied to a subsample of 25,000 spin configurations drawn randomly across all temperatures. Figure 4 shows the explained variance ratios for the first eight principal components. PC1 alone captures approximately 51% of the total variance, with a sharp drop to less than 1% for all subsequent components, as summarized in Table I. This single dominant eigenvalue is the first key result: it indicates that the data varies primarily along one direction in configuration space, which Wang [2] identifies as the signature of a system governed by a single dominant order parameter.

Principal Component	Explained Variance (%)
PC1	51.16
PC2	0.41
PC3	0.39
PC4	0.38
PC5	0.36
PC6	0.31
PC7	0.31
PC8	0.30

TABLE I. Explained variance ratios for the first eight principal components. PC1 dominates with 51.16%, consistent with a single dominant order parameter. The standard deviation of the PC1 weight vector is 5.63×10^{-5} , confirming it is spatially uniform.

The physical meaning of PC1 is confirmed by inspecting its weight vector, shown in Figure 5. The PC1 loading map is nearly spatially uniform across the lattice, with a standard deviation of 5.63×10^{-5} . Since the projection onto PC1 is the inner product of the weight vector with a spin configuration, this means $p_1 \approx \frac{1}{\sqrt{N}} \sum_i S_i$, which is exactly the total magnetization up to normalization. PCA has therefore au-

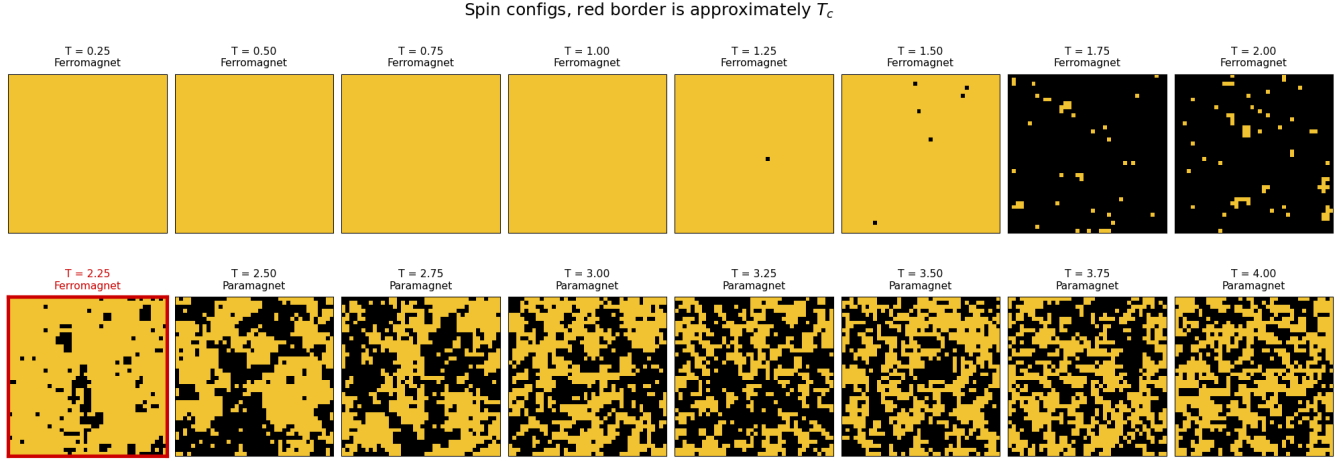


FIG. 1. Spin configurations of the 2D Ising model on a 40×40 lattice at temperatures ranging from $T = 0.25$ to $T = 4.00$. The ferromagnetic phase (uniform yellow) is visible from $T = 0.25$ to $T = 1.50$. Near $T_c \approx 2.25$ (red border), large fluctuating domains appear. Above T_c , the configurations become fully disordered, characteristic of the paramagnetic phase.

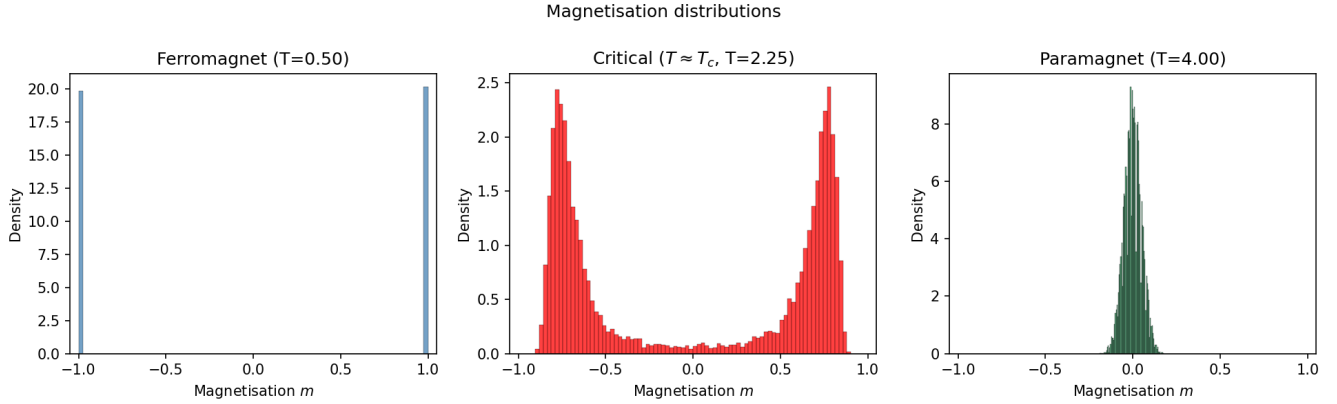


FIG. 2. Magnetisation distributions at three representative temperatures. At $T = 0.50$ (FM), the distribution is sharply peaked at $m \approx \pm 1$. At $T = 2.25$ (near T_c), it broadens into a bimodal structure reflecting critical fluctuations. At $T = 4.00$ (PM), it collapses to a single peak at $m \approx 0$.

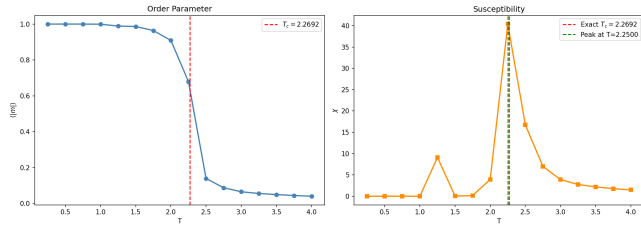


FIG. 3. Left: order parameter $\langle |m| \rangle$ as a function of temperature, dropping sharply near the exact $T_c = 2.2692$ (dashed red). Right: magnetic susceptibility χ , peaking at $T = 2.25$, within one temperature step of the exact value.

tonomously recovered the magnetization as the dominant order parameter, without any prior knowledge of the Hamilto-

nian or phase structure, directly reproducing the central finding of Wang [2].

Figure 6 shows the distribution of PC1 scores at three temperatures. At $T = 0.50$, the distribution is sharply peaked at $p_1 \approx \pm 40$, reflecting the nearly fully aligned configurations. At $T = 4.00$, it collapses to a single peak around zero, consistent with the disordered phase. Near T_c at $T = 2.25$, the distribution is broad and bimodal, corresponding directly to the large fluctuations at the critical point.

Figure 7 shows the temperature dependence of $\langle |p_1| \rangle$, $\langle |p_2| \rangle$, and $\langle |p_3| \rangle$, normalized and plotted alongside the true $\langle |m| \rangle$. The curve for $\langle |p_1| \rangle$ closely tracks $\langle |m| \rangle$ across the full temperature range, dropping sharply near T_c , with an estimated $T_c = 2.25$ from the steepest descent, in good agreement with the exact value of 2.2692. This confirms Wang's finding that PCA autonomously learns the magnetization as

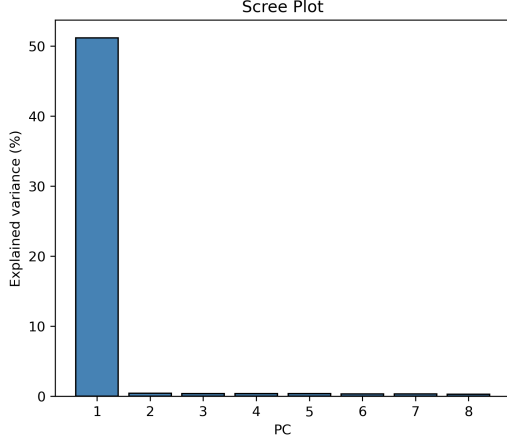


FIG. 4. Scree plot showing the explained variance ratio for the first eight principal components.

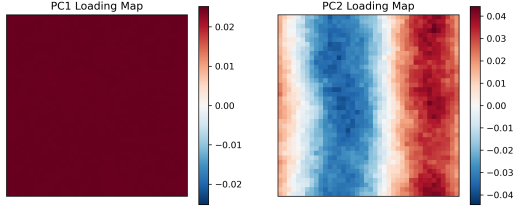


FIG. 5. PC1 (left) and PC2 (right) weight vectors reshaped to the 40×40 lattice.

the order parameter.

A notable observation is that PC2 and PC3 both show a pronounced peak at approximately $T = 2.50$, just above T_c , and their curves are nearly identical to each other throughout. This is consistent with Hu et al. [3], who show that the subleading principal components correspond to the next most

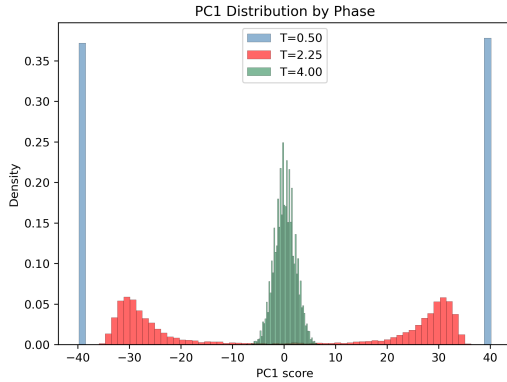


FIG. 6. Distribution of PC1 scores at three temperatures. At $T = 0.50$ (FM), the distribution is sharply peaked at $p_1 \approx \pm 40$. At $T = 2.25$ (near T_c), it is broad and bimodal. At $T = 4.00$ (PM), it collapses to a single peak near zero.

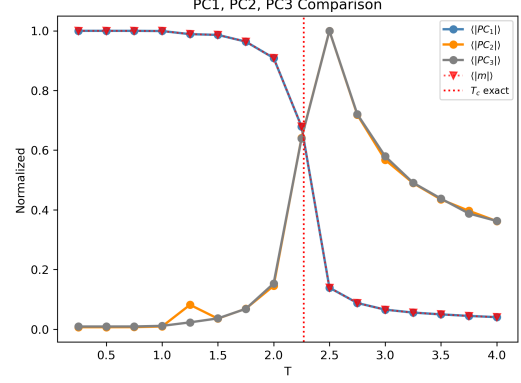


FIG. 7. Normalized $\langle p_1 \rangle$ (blue circle), $\langle p_2 \rangle$ (orange circle), and $\langle p_3 \rangle$ (gray circle) as functions of temperature, plotted alongside the true $\langle m \rangle$ (red triangles).

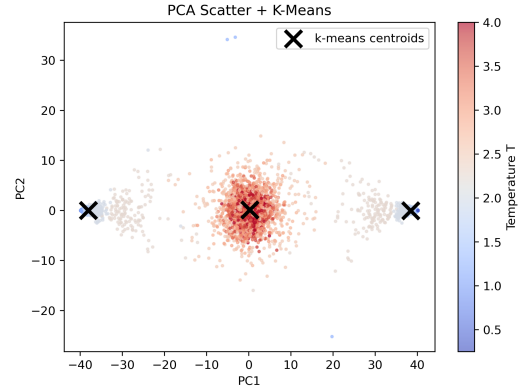


FIG. 8. PCA scatter plot in the PC1–PC2 plane coloured by temperature, with k -means centroids ($k = 3$, black crosses) overlaid.

ordered arrangements, specifically single horizontal or vertical domain wall modes. The PC2 weight map in Figure 5 supports this, showing a spatially modulated pattern consistent with the Fourier modes identified by Hu et al. [3].

C. k -Means Clustering

Figure 8 shows the PCA scatter plot in the PC1–PC2 plane with three k -means centroids overlaid. The algorithm places one centroid at the origin, corresponding to the paramagnetic cluster, and two symmetrically at $p_1 \approx \pm 40$, corresponding to the two ferromagnetic ground states. This three-cluster structure directly mirrors the finding of Wang [2], where the same method on a 40×40 lattice yields vertical decision boundaries in the PC1 direction. This is a crucial observation: it reinforces that the phase transition between FM and PM is captured entirely by the magnetization, and that the unsupervised clustering recovers the correct phase boundaries without any label information.

D. Autoencoder

Figure 9 shows the results of the autoencoder with a 2D latent space. The training loss (MSE) drops steeply and converges smoothly over 50 epochs. The latent space scatter plot colored by temperature shows that FM and PM configurations separate along the Z_1 axis, while Z_2 carries comparatively little information. When colored by phase label, the two FM ground states form two distinct clusters at large $|Z_1|$, separated by the paramagnetic cluster near $Z_1 \approx 0$, analogous to the three-cluster structure seen in PCA. The normalized $\langle |Z_1| \rangle$ curve closely tracks the true $\langle |m| \rangle$ as a function of temperature, with a T_c estimate consistent with 2.25. The per-temperature reconstruction error independently estimates $T_c = 2.25$, identified as the temperature of steepest rise in the MSE curve, consistent with the estimate from $\langle |Z_1| \rangle$ and the exact value. This confirms the findings of Hu et al. [3] and Wetzel [1] that the autoencoder latent coordinate spontaneously encodes the order parameter without any label supervision.

It should be noted that the autoencoder latent space is sensitive to training choices. The cluster structure shown in Figure 9 reflects a particular training sample size of 25,000 configurations and 50 epochs; varying these parameters produces qualitatively similar but quantitatively different latent embeddings. However, while the precise embedding may differ, Z_1 is expected to consistently separate the FM and PM phases. This sensitivity is an intrinsic limitation of standard autoencoders compared to PCA, which is deterministic given the same data subsample, and motivates the use of a Variational Autoencoder.

E. Variational Autoencoder

Figure 10 shows the full VAE analysis. The training loss converges within approximately 10 epochs. The Kullback–Leibler divergence term remains near zero throughout training, indicating that the 1D latent space is sufficient for the encoder to represent the data without over-regularization.

The central panel shows the mean reconstruction error (MSE) as a function of temperature. The error is near zero in the deep FM phase, rises sharply near T_c , and peaks at $T = 2.25$, in close agreement with the exact value of 2.2692. This confirms the result of Wetzel [1] that the reconstruction error serves as a universal identifier for phase transitions: the VAE reconstructs ordered configurations faithfully, but fails near and above T_c because disordered configurations have no common structure for the network to exploit.

The latent scatter plot shows that the VAE posterior mean μ is tightly linearly correlated with the true magnetisation m , closely reproducing the result shown by Wetzel [1]. The histogram of μ values displays three clusters: peaks at $\mu \approx \pm 2$ corresponding to the two FM ground states, and a broad central distribution at $\mu \approx 0$ corresponding to the PM phase. This three-cluster structure in a 1D latent space is consider-

ably cleaner and more interpretable than the AE latent space, where the second dimension Z_2 adds no physical information. We note that because the VAE uses the reparameterisation trick during training but returns μ directly at inference time, μ acts as a deterministic and stable order parameter proxy.

The right panel compares $\langle |\mu| \rangle$ from the VAE, $\langle |Z_1| \rangle$ from the AE, and the true $\langle |m| \rangle$. All three track each other closely, though the VAE curve is marginally smoother near T_c . The main advantage of the VAE, as argued by Wetzel [1], is that the Gaussian prior on the latent space regularises the encoder and forces μ to carry physically meaningful information, making it more robust than a standard AE particularly near T_c where the sign of m is not fixed.

Figure 11 shows VAE reconstructions at three temperatures. At $T = 0.50$, the reconstruction is nearly perfect, with the network correctly outputting a uniform spin configuration. At $T = 2.25$, the reconstruction becomes blurry: the VAE encodes the configuration at an intermediate μ value and the decoder outputs the mean spin value, unable to resolve the fine domain structure. At $T = 4.00$, the reconstruction is grey, corresponding to the decoder outputting the mean of the prior at $\mu = 0$. This behaviour is a direct consequence of the MSE loss, which is minimised by predicting the mean rather than any specific disordered pattern.

F. t-SNE

Inspired by Wetzel [1], Figure 12 shows the two-component t-SNE embedding of 5,000 configurations, colored first by temperature and then by phase label. The embedding reveals three distinct clusters: two outer clusters corresponding to the two FM ground states ($m \approx +1$ and $m \approx -1$), and a central cluster corresponding to the PM phase. When colored by phase label, some red points appear near the boundary of the central cluster, which is expected as these correspond to configurations near T_c where the phase boundary is not sharp. This three-cluster structure is consistent with the two-component t-SNE results of Wetzel [1], where ordered configurations map to the exterior of the embedding and disordered configurations to the interior. The separation of the two FM ground states into distinct outer clusters arises because t-SNE preserves local distances: configurations with $m \approx +1$ are far from those with $m \approx -1$ in spin space and are therefore placed in separate clusters, consistent with the left-right symmetry also observed by Wetzel.

G. Supervised Baseline: CNN

Finally, a convolutional neural network (CNN) is applied as a supervised baseline. The CNN achieves a validation accuracy above 99% within two epochs, with the training loss dropping rapidly and converging near zero, as shown in Figure 13. T_c is estimated as 2.25 from the crossing of the predicted ferromagnetic probability, consistent with the ex-

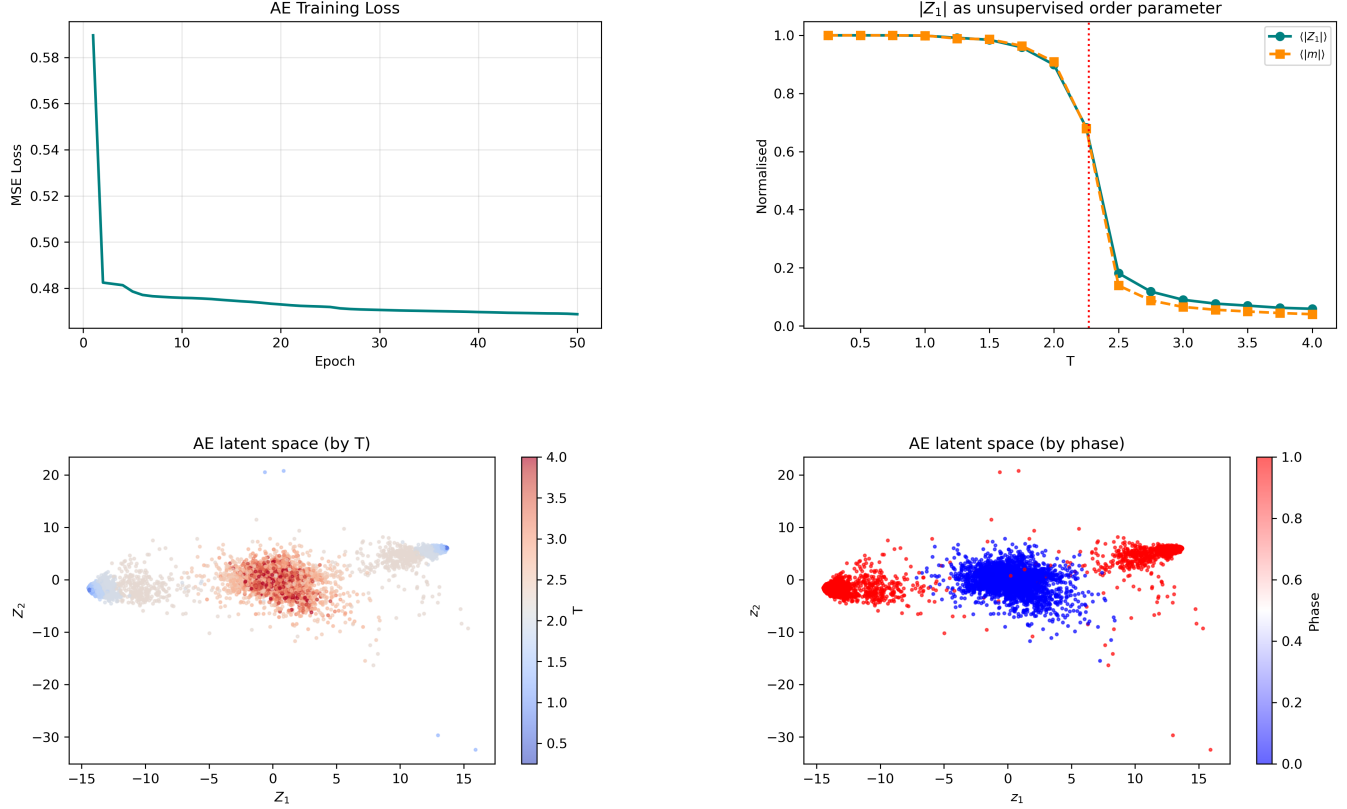


FIG. 9. Autoencoder results. Top left: training loss (MSE) over 50 epochs. Top right: $\langle |Z_1| \rangle$ compared to true $\langle |m| \rangle$, estimating $T_c = 2.25$. Bottom left: latent space colored by temperature, showing FM-PM separation along Z_1 . Bottom right: latent space colored by phase label.

act value. This strong performance is expected: with clean phase labels and a convolutional architecture suited to spatial structure, classification of FM versus PM is a straightforward task. The CNN result serves primarily to confirm that the data is clean and that T_c can be accurately located from labeled configurations, providing a ceiling against which the unsupervised methods can be judged. Notably, all unsupervised methods recover $T_c = 2.25$, within one temperature step of the exact value and consistent with the supervised result, demonstrating that labeled data is not necessary for locating the phase transition in this model.

IV. DISCUSSION AND CONCLUSION

This paper applied a suite of unsupervised and supervised machine learning methods to Monte Carlo configurations of the 2D ferromagnetic Ising model on a 40×40 lattice, with the goal of identifying the phase transition and recovering the order parameter without prior physical knowledge. The central finding is that unsupervised methods — PCA, the autoencoder, and the variational autoencoder — are each capable of independently locating the critical temperature and extracting physically meaningful representations of the order parameter

directly from raw spin configurations. All methods consistently recover $T_c = 2.25$, within one temperature step of Onsager’s exact value of 2.2692, and in agreement with the results of Wang [2], Wetzel [1], and Hu et al. [3], whose central findings are reproduced here.

Among the unsupervised methods, PCA is the most interpretable and computationally efficient: it autonomously recovers the magnetization as the dominant order parameter and produces a deterministic result given the same data subsample. The VAE produces a cleaner and more stable latent representation than the standard autoencoder, particularly near T_c , due to the Gaussian prior regularizing the latent space. The standard autoencoder, while effective, is sensitive to training choices and introduces stochasticity that PCA avoids. The supervised CNN achieves the highest classification accuracy but requires labeled data and provides no physical interpretation beyond locating T_c . Together, these results confirm that labeled data is not necessary for detecting phase transitions in this model.

However, this paper has several important limitations that constrain the generality of these conclusions. First, all experiments are conducted on a single lattice size of $L = 40$. It is not established here whether the same methods would perform comparably on smaller lattices such as $L = 20$ or $L = 30$, where finite-size effects are more pronounced and

Variational Autoencoder (VAE) Analysis

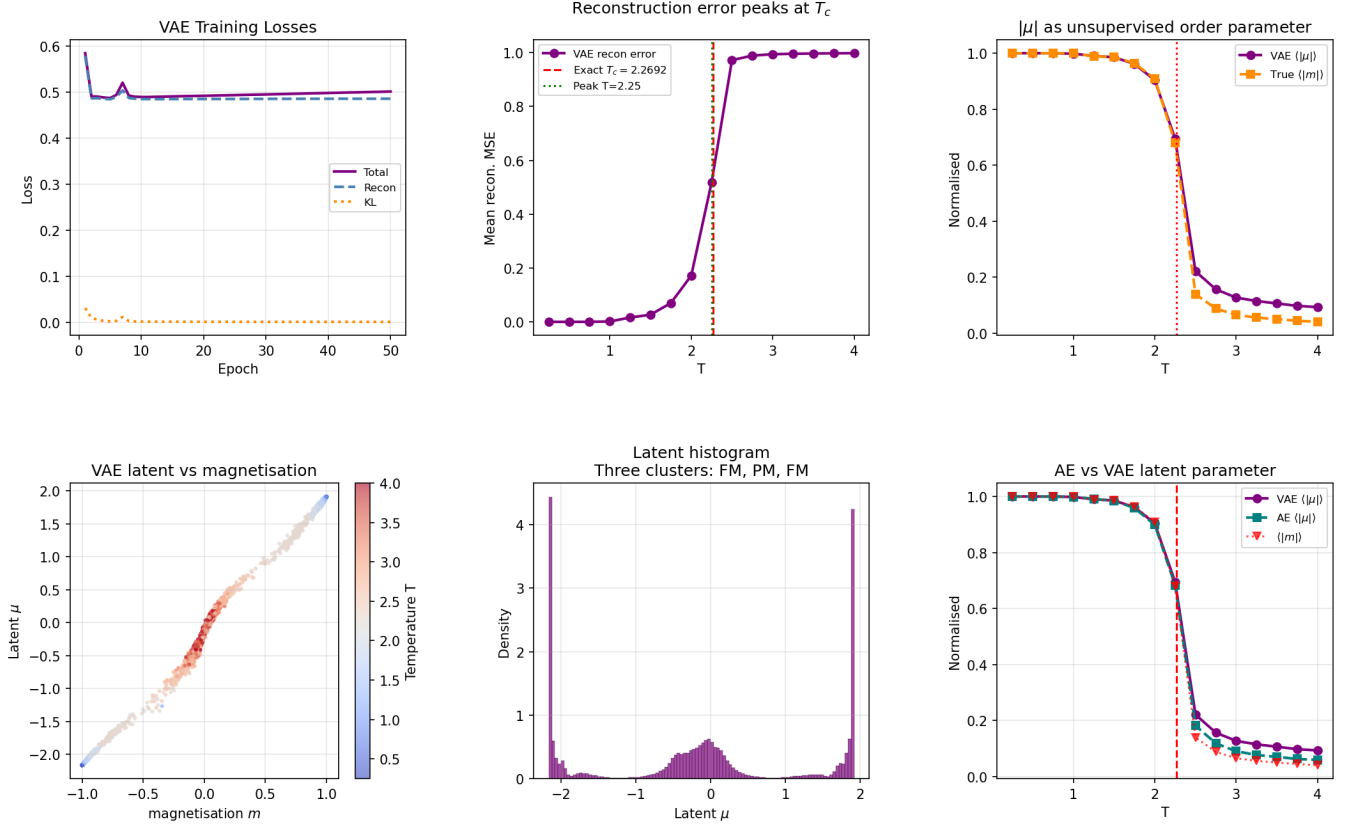


FIG. 10. VAE analysis. Top left: training loss components (total, reconstruction MSE, KL divergence) over 50 epochs. Top center: mean reconstruction error as a function of temperature, peaking at $T = 2.25$. Top right: normalized $\langle \mu \rangle$ compared to true $\langle m \rangle$. Bottom left: latent posterior mean μ versus true magnetization m , showing tight linear correlation. Bottom center: histogram of μ values, showing three clusters corresponding to the two FM ground states ($\mu \approx \pm 2$) and the PM phase ($\mu \approx 0$). Bottom right: comparison of $\langle \mu \rangle$ (VAE), $\langle Z_1 \rangle$ (AE), and $\langle m \rangle$.

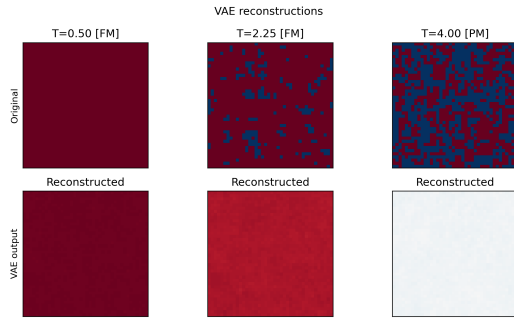


FIG. 11. VAE spin configuration reconstructions at three temperatures.

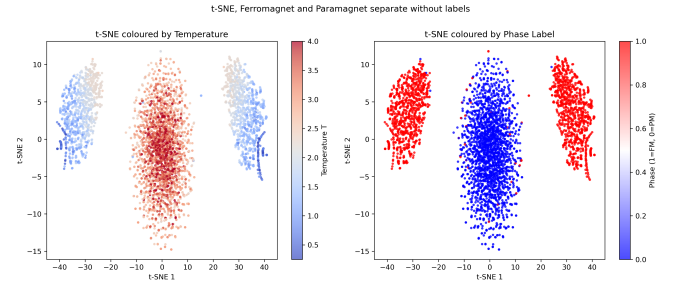


FIG. 12. Two-component t-SNE embedding of 5,000 configurations. Left: coloured by temperature. Right: coloured by phase label (red = FM, blue = PM)

the transition is less sharp. A natural extension would be to parameterize the data pipeline by L and systematically test each method across multiple lattice sizes, which would also

allow a finite-size scaling analysis to obtain a more precise estimate of T_c .

Second, the temperature grid is discrete with a step size of

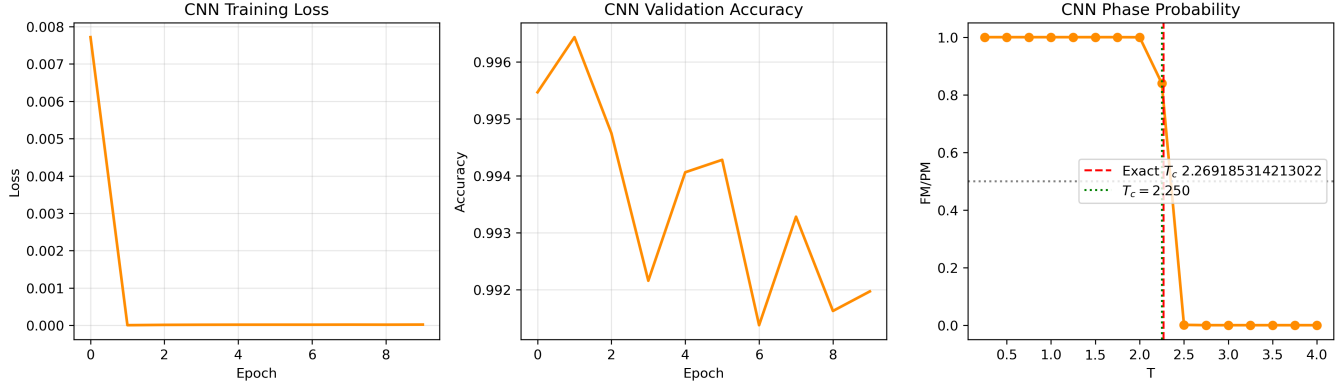


FIG. 13. CNN supervised baseline results. Left: training loss over 10 epochs, converging near zero. Center: validation accuracy. Right: mean predicted ferromagnetic probability $P(\text{FM})$ as a function of temperature.

$\Delta T = 0.25$. A finer temperature grid, or a Monte Carlo simulation that samples more densely near the suspected critical region, would allow a more precise determination of T_c and a cleaner characterization of the order parameter curve near the transition.

Third, and most importantly, the 2D ferromagnetic Ising model has a simple, linear order parameter — the magnetization — which is precisely the setting in which PCA and linear methods are expected to perform well. None of the methods have been tested here on systems with nonlinear or topological order parameters, such as the systems originally motivating this line of research [1, 2]. Extending the Monte Carlo simulations to generate configurations from such systems and

re-running the same pipeline would be a critical next step in assessing whether the unsupervised approach generalizes beyond the Ising universality class.

In summary, this paper demonstrates as a controlled proof of concept that unsupervised machine learning methods can recover both the order parameter and the critical temperature of the 2D Ising model without any label supervision, reproducing and confirming the results of the key literature [1–3]. The primary value of this verification is as a foundation: the same pipeline, with broader data generation across multiple lattice sizes and extended testing on nonlinear systems, could in principle be applied to systems where no exact solution exists and the order parameter is not known in advance.

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